

KISANLINK

A Farmer-Centric Procurement Slot, Queue and Status Management Platform

Research Paper / Proposed System Study

PROJECT FOCUS

Making the government procurement journey more predictable through centre discovery, scheduling, queue visibility, notifications and status tracking.

Presented by

SYNTAX ERROR

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Agricultural Procurement

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Prepared for academic and project presentation purposes

Abstract

Government procurement at Minimum Support Price (MSP) is an important component of India's agricultural policy. The procurement ecosystem has already undergone significant digitization through central and state procurement portals, the Central Foodgrains Procurement Portal (CFPP), e-NAM for agricultural market trading, and agency-specific platforms such as NAFED's e-Samridhi and NCCF's e-Samyukti. However, evidence from research and field reporting shows that farmers can still face operational uncertainty at the last mile: delayed or irregular procurement windows, long waits or serial-number queues, transport and infrastructure constraints, documentation/registration issues, quality-related rejection, and uncertainty about the next step. This paper proposes KisanLink, a farmer-centric coordination layer designed to make the procurement-centre journey more predictable. The proposed system provides state-to-district-to-centre discovery, farmer onboarding, eligibility-aware centre selection, slot/token management where supported, queue and estimated-wait visibility, notifications, procurement-status tracking, and an operator dashboard. An AI layer is proposed for demand forecasting, queue-pressure estimation, congestion detection and load balancing. The paper deliberately positions KisanLink as a complementary orchestration platform rather than a replacement for government procurement systems. The proposed approach can be evaluated through measurable indicators such as waiting time, unnecessary trips, slot adherence, centre utilization, status visibility and farmer satisfaction.

Keywords: agricultural procurement, MSP, farmers, procurement centres, queue management, digital agriculture, slot booking, farmer services, artificial intelligence, e-governance, FCI, NAFED, NCCF, CFPP.

Research status: This is a proposed-system research paper. KisanLink performance values are not presented as measured results; they are proposed evaluation metrics for a future pilot.

1. Introduction

Agricultural procurement is not simply a transaction between a farmer and a buyer. In government procurement, the process can involve farmer registration, land and crop verification, centre allocation or selection, arrival scheduling, quality assessment, weighing, acceptance, storage and payment. The Department of Food and Public Distribution (DFPD) states that FCI and State Government agencies procure wheat and paddy at MSP for the Central Pool under centralized and decentralized procurement arrangements. Procurement plans are prepared with regard to production, marketable surplus and crop patterns. [1]

The scale of this ecosystem makes operational coordination important. Government documents describe online farmer registration, integration with land records, digitized procurement-centre operations, online MSP payment and API-based reporting to CFPP as components of the digital procurement architecture. [2][3] Therefore, the contemporary problem is not accurately described as an absence of technology.

The more relevant research question is whether the farmer's end-to-end procurement journey is sufficiently predictable and transparent. A farmer may have successfully registered and still face uncertainty about the appropriate centre, procurement window, current queue, operational capacity, arrival timing, quality-related processing or transaction status. Research from different states provides evidence of procurement delays, long waits, transport and infrastructure problems, documentation/registration barriers and other local operational constraints. [4][5][6][7]

KisanLink is proposed in response to this coordination gap. The system is designed as a farmer-facing orchestration layer that can consume authorized procurement information and convert it into a simple decision journey: *where should I go, when should I go, what is the expected wait, what is my current status, and what should I do next?*

Central research proposition: If verified procurement-centre information, appointment/queue mechanisms and transaction-status events are presented through a unified farmer-centric interface, then unnecessary uncertainty and avoidable congestion can be reduced without replacing the underlying government procurement agencies or portals.

2. Background and Procurement Ecosystem

2.1 Government procurement at MSP

Under the DFPD procurement policy, wheat and paddy are procured by FCI and State Government agencies at MSP, using centralized and decentralized procurement arrangements. In centralized procurement, FCI or state agencies procure for the Central Pool; in decentralized procurement, the State Government undertakes direct purchase, storage and distribution within the applicable framework. [1]

2.2 Digital procurement architecture

Government procurement has already adopted multiple digital components. FCI's procurement action plan describes online registration, land-record integration, digitized mandi/procurement-centre operations, online payment and API-based reporting to CFPP. [2] The NIC description of CFPP similarly identifies online farmer registration, farmer land records, digitized procurement-centre operations, MSP payment and API integration as threshold parameters for procurement applications. [3]

2.3 Other digital agricultural platforms

e-NAM is a pan-India electronic trading portal that networks existing APMC mandis to facilitate agricultural commodity trading and price discovery. Its farmer module allows farmers to register, view lots, monitor lot progress and access market information. e-NAM therefore addresses electronic market trading rather than being a dedicated government procurement-centre queue-management layer. [8][9]

For procurement under the Price Support Scheme, government material describes e-Samridhi, developed by NAFED, and e-Samyukti, developed by NCCF, as platforms that streamline processes from farmer registration to final payment. The same source states that registered farmers can choose a nearby purchase centre and receive digitally scheduled slots for visits. [10] This is an important finding for KisanLink because it means slot scheduling itself cannot be claimed as a completely new invention.

Consequently, KisanLink's innovation should be framed around **unified farmer-facing coordination, centre discovery, queue/capacity visibility, notifications, status orchestration and analytics across a fragmented ecosystem**, subject to government data access and integration permissions.

3. Problem Statement

Farmers participating in government procurement may encounter uncertainty and operational delays even after entering a digital procurement workflow. The problem can be expressed as follows:

Farmers lack a single, understandable and timely view of the procurement-centre journey, including the appropriate centre, expected arrival time, queue/capacity situation and post-arrival status, while procurement-centre operators lack a unified farmer-arrival and congestion-management layer.

3.1 Evidence from research

A West Bengal study of decentralized paddy procurement reports long waiting periods for serial numbers at procurement centres, delays in procurement, intermediary interference and other process difficulties. It also reports substantial levels of delayed procurement among sampled small and marginal farmers. [4] These findings are geographically specific and should not be treated as a national prevalence estimate.

A study of farmer participation in West Bengal's government e-procurement found that knowledge of the procurement system, registration status and distance from the market/mandi were significant factors in the farmer's choice of market channel. [5] This supports the importance of information, accessibility and onboarding rather than assuming that a digital portal alone solves participation barriers.

A 2024 study of paddy procurement in Kerala, based on 160 farmers in selected Panchayats of Palakkad and Alappuzha, ranked lack of immediate payment as the most significant constraint, followed by difficulty in meeting quality standards. [6] This demonstrates that a queue-management product must not overclaim that every procurement problem is caused by queues; KisanLink should instead expose and track relevant stages.

Recent field reporting from Telangana also describes farmers waiting at procurement centres amid shortages or constraints involving sheds, drinking water, weighing equipment, transport and other logistics. [7] Such evidence highlights the operational conditions surrounding the digital transaction.

A research report on paddy procurement in Bihar and Odisha identifies irregular procurement windows, delayed payments, local influence, trader participation and incorrect data as critical bottlenecks. It is especially relevant to KisanLink because it shows that real-time tracking technology can coexist with last-mile operational gaps. [11]

4. Research Gap and Rationale

The literature and official ecosystem together reveal a specific gap. Government systems increasingly support registration, verification, procurement recording and payment, while e-NAM provides a separate digital trading environment. Agency-specific systems may also provide scheduling. Yet the farmer often experiences these functions as a sequence of operational decisions rather than as one coherent journey.

Existing capability	Current ecosystem evidence	Remaining opportunity for KisanLink
Farmer registration	State and agency systems support online registration and identity/land/crop information.	Unified farmer profile and clear pre-visit checklist, subject to scheme rules.
Centre selection	Agency platforms can allow farmers to choose purchase centres.	State → District → Centre discovery with operational status and contextual information.
Digital scheduling	e-Samridhi/e-Samyukti already describe digital slot scheduling for relevant procurement operations.	Cross-system farmer experience and transparent scheduling/queue journey rather than claiming scheduling is new.
Procurement monitoring	CFPP and state portals support reporting and monitoring.	Farmer-readable status timeline and actionable notifications.
Payment	Direct digital MSP payments are already supported in multiple systems.	Payment/status visibility and exception guidance; not a replacement for payment infrastructure.
Market information	e-NAM provides prices, lots and electronic trading functions.	Focus specifically on government procurement-centre coordination, not general mandi trading.

Novelty claim recommended for the project: KisanLink is a proposed farmer-centric orchestration and queue-management layer for government procurement-centre journeys, combining centre discovery, capacity/queue visibility, scheduling where supported, notifications, transaction-status presentation and analytics in one interface.

5. Objectives

- **O1:** Reduce uncertainty before a farmer travels to a procurement centre.
- **O2:** Provide structured discovery of procurement centres by state, district, crop and relevant agency/process.
- **O3:** Support digital slots or tokens where the underlying procurement process permits them.
- **O4:** Present queue position or estimated waiting range using available operational data.
- **O5:** Provide timely notifications for appointments, centre-status changes and transaction events.
- **O6:** Present the procurement journey as a simple status timeline.
- **O7:** Give centre operators visibility into arrivals, workload, pending cases and congestion.
- **O8:** Use AI only for tasks where data can support meaningful prediction, such as demand forecasting and queue-pressure estimation.
- **O9:** Support multilingual and assisted-access pathways for farmers with limited digital literacy or connectivity.
- **O10:** Define measurable pilot KPIs rather than claiming unverified percentage improvements.

6. Proposed KisanLink System

6.1 Functional modules

Module	Primary function	Primary user
Farmer Onboarding	Profile, identity, land/crop, bank and scheme-related details as required.	Farmer
Centre Directory	State → District → Centre classification, location, crop/process, operating information.	Farmer / Operator
Recommendation Engine	Suggest suitable centres using eligibility, distance and capacity signals.	Farmer
Slot/Token Engine	Manage available appointment/token windows where supported.	Farmer / Operator
Queue Engine	Track arrivals and estimate queue pressure/waiting range.	Farmer / Operator
Notification Service	Send reminders, changes, queue and status alerts.	Farmer
Procurement Status	Display registered → scheduled → arrived → quality → weighing → accepted/rejected → payment status.	Farmer
Operator Dashboard	Monitor daily workload, arrivals, pending cases and congestion.	Centre Operator
Analytics & AI	Forecast demand, identify anomalies and support load balancing.	Operator / Administrator
Admin & Governance	Data permissions, centre configuration, audit logs and access control.	Administrator

6.2 High-level architecture

Farmer / Operator Interface → Application API → Business Services → Authorized Data Sources

The business services may include identity/profile, centre directory, scheduling, queue management, notifications, procurement-status and analytics modules. Authorized data sources may include state procurement portals, agency systems, centre operator inputs and approved APIs. The architecture should avoid scraping or fabricating government data in a production deployment.

Prototype principle: If official APIs are unavailable during a student prototype, seeded or simulated data may be used to demonstrate the workflow. The UI should explicitly label such data as demonstration data rather than implying a live government integration.

7. Proposed Farmer Workflow

Stage	Farmer action	System response
1. Registration	Creates a complete profile.	Validates required fields and creates farmer record.
2. Procurement intent	Selects crop and procurement requirement.	Maps request to applicable process/agency rules where configured.
3. Centre discovery	Selects state and district or uses location.	Shows relevant centres and operational information.

Stage	Farmer action	System response
4. Centre choice	Reviews distance, status and available capacity.	Ranks centres using configurable rules.
5. Slot/token	Selects available slot/token where supported.	Confirms appointment and generates reference.
6. Pre-arrival	Waits for assigned window or alert.	Sends reminder and changes if centre status changes.
7. Check-in	Arrives and checks in.	Updates queue/arrival state.
8. Processing	Completes quality and weighing steps.	Updates status timeline.
9. Completion	Views accepted quantity and payment/settlement status.	Displays authorized transaction status.
10. Feedback	Reports issue or rates service.	Creates feedback/issue record for analysis.

This workflow separates **pre-arrival uncertainty** from **on-site processing**. KisanLink can reduce the first category directly and can improve the second through better arrival distribution and operator visibility. It cannot, by itself, eliminate physical constraints such as lack of storage, labour, trucks, weighing equipment or weather-related disruptions.

8. Artificial Intelligence Component

AI is proposed as a decision-support layer rather than a marketing label. The system should use predictive methods only when historical and operational data are sufficient and authorized. If the student prototype does not have real procurement data, synthetic data can be used to demonstrate the method.

AI use case	Potential inputs	Output	Operational value
Demand forecasting	Historical arrivals, crop, season, weekday, centre capacity.	Expected arrivals by time window.	Improved slot/capacity planning.
Queue estimation	Current queue, service rate, appointments, centre status.	Estimated waiting range.	Farmer can plan arrival more effectively.
Load balancing	Nearby centres, eligibility, distance, capacity.	Alternative centre suggestion.	Reduces concentration at overloaded centres.
Anomaly detection	Historical processing patterns vs current observations.	Congestion/slowdown alert.	Earlier operator intervention.
Notification intelligence	Appointment and centre events.	Relevant alert timing.	Reduces missed appointments and unnecessary travel.

Important limitation: Queue predictions are estimates, not guarantees. Weather, equipment failure, quality rejection, transport availability, power/network outages and staffing can change actual processing time. The interface should communicate uncertainty rather than display a false precise guarantee.

9. Data and Technology Requirements

9.1 Core entities

Entity	Illustrative fields
Farmer	Farmer ID, name, mobile, address, language preference, identity reference, bank/payment reference, land/crop information.
Procurement Centre	Centre ID, state, district, location, crops/processes, agency, operating window, capacity, contact, status.
Slot/Token	Slot ID, centre ID, date, time window, capacity, booked count, status.
Queue Event	Check-in time, token, queue position, service start, service completion, exception.
Procurement Transaction	Farmer ID, crop, quantity, quality result, acceptance, transaction reference, payment status.
Notification	Recipient, event type, message, timestamp, delivery state.
Operator	Role, centre assignment, permissions, audit information.

9.2 Privacy and security

- Collect only information required for the selected procurement workflow.
- Use role-based access control so farmers, centre operators and administrators see only relevant information.

- Protect identity and bank-related information using appropriate encryption and secure credential management.
- Maintain audit logs for changes to slots, queue states and transaction-status data.
- Use consent and lawful authorization requirements applicable to the final deployment.
- Do not expose Aadhaar, bank account or other sensitive identifiers unnecessarily in the user interface.

10. Evaluation Methodology

Because KisanLink is a proposed platform, a credible research evaluation should compare a baseline procurement-centre process with a KisanLink-assisted process in a controlled pilot or simulation. The study should not claim improvement before measurements are collected.

Metric	Definition	Expected direction
Average waiting time	Time between farmer check-in and processing start.	Decrease
Total visit duration	Arrival/check-in to completion of procurement interaction.	Decrease
Unnecessary trips	Visits where the farmer could not be processed due to closure, capacity or scheduling mismatch.	Decrease
Slot adherence	Share of farmers arriving within their assigned window.	Increase
Centre utilization	Actual processing workload relative to planned capacity.	More balanced
Status visibility	Share of farmers able to identify their current stage without manual enquiry.	Increase
Notification effectiveness	Delivery and acknowledgement of relevant alerts.	Increase
Farmer satisfaction	Post-visit rating or structured survey score.	Increase
Operator workload	Manual coordination actions required per farmer/day.	Decrease

10.1 Suggested pilot design

Select a small number of procurement centres with comparable operating conditions. Measure a baseline period, deploy the KisanLink workflow, and compare the same metrics during the intervention period. Record confounders such as rainfall, crop arrival peaks, equipment outages, holidays and staffing changes. A randomized or matched-centre design would provide stronger evidence than a simple before/after comparison.

11. Comparative Positioning

Capability	Government / existing platform examples	KisanLink position
MSP procurement	FCI and State agencies under applicable procurement policy.	Complementary coordination layer.
Foodgrain procurement reporting	CFPP and state procurement systems.	Consume authorized information; do not duplicate unnecessarily.
Electronic agricultural trading	e-NAM.	Not a replacement; KisanLink focuses on procurement-centre operations.
Agency procurement	NAFED e-Samridhi, NCCF e-Samyukti and other agency systems.	Unified farmer journey across relevant services where integration is authorized.
Digital scheduling	Already available in some agency procurement workflows.	Use where supported; differentiate through unified queue/status/centre experience.
AI-based forecasting	Not assumed to be a universal existing feature across all systems.	Proposed predictive layer for demand and congestion.
Centre-level queue visibility	May vary by state/agency and implementation.	Core KisanLink focus.

The comparison demonstrates why the project can be credible without claiming that it is the first digital procurement platform. Its contribution is integration of the farmer journey and operational visibility.

12. Expected Benefits

12.1 Farmer-level benefits

- Lower uncertainty before travelling to a procurement centre.
- Better visibility of centre location, operating information and appointment/token status.
- Potential reduction in unnecessary waiting and repeat visits.
- Simpler understanding of procurement status and next action.
- Multilingual and assisted-access pathways can improve usability.

12.2 Centre-level benefits

- Better visibility of expected arrivals and workload.
- Earlier detection of congestion and abnormal processing slowdowns.
- Potentially smoother distribution of arrivals across available capacity.
- Digital record of queue and operational events for analysis.

12.3 Administrative benefits

- Aggregated analytics on centre utilization and recurring bottlenecks.
- More structured feedback from farmers.
- Potential interoperability with authorized government systems rather than creating another isolated database.

13. Limitations and Risks

- **Data availability:** Real-time queue prediction requires reliable and timely operational data.
- **Integration dependency:** Production deployment depends on APIs, data-sharing permissions and government/agency cooperation.
- **Physical constraints:** Software cannot directly remove shortages of trucks, labour, storage, weighing equipment or shelter.
- **Digital divide:** A smartphone-only solution may exclude some farmers; assisted channels are necessary for inclusive deployment.
- **Prediction uncertainty:** AI estimates can be wrong during unusual events and should be shown with appropriate uncertainty.
- **Privacy:** Farmer identity, land and payment information require strong security and governance.
- **Scheme variability:** Procurement rules, eligible crops, agencies and centre processes vary by state, season and scheme.
- **Prototype limitations:** Student demonstrations using mock data must not be represented as live government data.

14. Future Work

- Develop standardized adapters for state and agency procurement systems.
- Integrate verified centre-status and operational-capacity feeds.
- Introduce SMS/IVR/CSC/PACS-assisted workflows for low-connectivity users.
- Train forecasting models on anonymized historical arrival and processing data.
- Add multilingual voice assistance for farmers.
- Conduct field pilots and statistically evaluate reductions in waiting time and unnecessary trips.
- Develop governance standards for data sharing, access control, auditability and model monitoring.
- Extend the centre directory to cover relevant procurement agencies and crop-specific procurement processes across states.

15. Conclusion

Government agricultural procurement in India is increasingly digital. Official sources document online registration, land-record integration, digital procurement-centre operations, direct MSP payments and centralized reporting. Agency-specific systems such as e-Samriddhi and e-Samyukti also provide digital registration, centre selection and scheduling for relevant procurement programs. [2][3][10] The existence of these systems is not a weakness for KisanLink; it defines the problem more precisely.

The research indicates that procurement challenges persist at the operational and last-mile level. Studies and field reports describe delayed procurement, long waits, transport and infrastructure constraints, payment delays, quality-related difficulties and information/registration barriers in specific contexts. [4][5][6][7][11] These findings support a farmer-centric coordination approach, while also showing that queue management alone cannot solve every procurement problem.

KisanLink therefore proposes a complementary architecture: a single farmer-facing interface that organizes procurement-centre discovery, scheduling where supported, queue visibility, notifications and procurement status, while giving centre operators workload and congestion visibility. AI is positioned as a decision-support component for forecasting and anomaly detection rather than as an unsupported promise of automatic problem solving.

Final proposition: KisanLink aims to transform the farmer's procurement experience from an uncertain physical visit into a more predictable, transparent and trackable digital journey — while remaining interoperable with, rather than replacing, the government systems that already perform core procurement functions.

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Appendix A. Research-to-Presentation Takeaways

Problem: The challenge is not that government procurement is completely offline. The stronger problem is fragmented, uncertain and sometimes operationally difficult last-mile procurement experience.

Evidence: Research and field reporting identify delays, waiting, centre-level constraints, transport/logistics problems, payment delays, quality issues and information/registration barriers in different contexts.

Existing systems: FCI/State portals, CFPP, e-NAM and agency platforms already solve important parts of the process.

Gap: The farmer needs one understandable journey across centre discovery, scheduling, queue/status visibility and next-step communication.

KisanLink: A complementary orchestration and queue-management platform, not a replacement for FCI, NAFED, NCCF or state procurement systems.

AI: Demand forecasting, queue-pressure estimation, load balancing and anomaly detection using authorized operational data.

Evaluation: Measure waiting time, unnecessary trips, slot adherence, centre utilization, status visibility, operator workload and farmer satisfaction.

One-line positioning: “KisanLink does not replace government procurement; it makes the farmer’s procurement journey more predictable.”

Important: Before submitting the final PPT, verify any current centre list, procurement schedule, agency rule, live statistic or API availability against the relevant official source. The system concept is general, but actual procurement rules are state-, crop-, agency- and season-dependent.